Stroma World — Reality Anchor Metrics & Pass/Fail Tests v1.0

> **Rule:** Every test is directional or relative. No absolute thresholds. If it requires a specific number to pass, it's a calibration target, not a reality anchor test. **Convention:** A > B means "A is measurably greater than B in the same simulation units, by any nonzero amount that exceeds stochastic noise across 3+ replicate runs."

Anchor 1: The Barrier Self-Assembles

Metrics

ID	Metric	How to measure
1.1	**CAF activation fraction**	Count of ACTA2=ON stromal cells / total stromal cells at endpoint
1.2	**ECM density gradient**	Mean ECM density in the 50μm shell adjacent to tumor vs. mean ECM density at the domain boundary
1.3	**Time to first CAF activation**	Simulated time from t=0 until the first PSC transitions to ACTA2=ON

Pass/Fail

Test	Criterion
P1a	CAF activation fraction at endpoint > CAF activation fraction at t=0 (which is 0). At least *some* PSCs must activate.

Test	Criterion
P1b	ECM density near tumor > ECM density at domain edge. The gradient must point inward, not be flat or inverted.
P1c	Time to first CAF activation < total simulation duration. Activation must happen spontaneously, not require the full run to barely trigger.
FAIL if	Any of: no PSCs activate, ECM is uniform or higher at domain edge than near tumor, or activation only occurs in the final 10% of simulation time (suggesting marginal/accidental triggering).

Anchor 2: Drug Diffusion Depth Drops With Barrier Maturity

Metrics

ID	Metric	How to measure
2.1	**Drug concentration at tumor centroid**	Sample drug substrate value at the geometric center of the tumor cell population
2.2	**Drug penetration half-depth**	Distance from domain boundary at which drug concentration drops to 50% of boundary value
2.3	**Barrier maturity index**	Mean ECM density in the peri-tumoral shell (proxy for barrier stage)

Measure all three at three time points: early (barrier immature), mid (barrier forming), late (barrier mature).

Pass/Fail

Test	Criterion
P2a	Drug at centroid (late) < Drug at centroid (mid) < Drug at centroid (early). Strict monotonic decline across all three time points.
P2b	Drug penetration half-depth (late) < Drug penetration half-depth (early). The drug front retreats as the barrier thickens.
P2c	Barrier maturity index increases monotonically across the three time points (confirming that the barrier is actually maturing, not that something else is consuming the drug).
FAIL if	Drug at centroid is constant or *increases* over time, or if half-depth doesn't shorten, or if barrier maturity index doesn't increase (meaning the test conditions weren't met).

Anchor 3: SHH Inhibition Alone Worsens Outcome

Metrics

ID	Metric	How to measure
3.1	**Total tumor cell count**	Number of viable tumor cells at endpoint
3.2	**Tumor spatial extent**	Maximum radius from tumor centroid to outermost tumor cell
3.3	**Barrier density**	Mean ECM density in the peri-tumoral shell

Compare: **Control** (no intervention) vs. **SHH-blocked** (shh_secretion_rate = 0, no drug).

Pass/Fail

Test	Criterion
P3a	Tumor cell count (SHH-blocked) > Tumor cell count (Control). Removing SHH makes the tumor bigger, not smaller.
P3b	Tumor spatial extent (SHH-blocked) > Tumor spatial extent (Control). The tumor physically expands more without the SHH-driven stroma component.
P3c	Barrier density (SHH-blocked) < Barrier density (Control). Confirms the intervention actually reduced stroma (otherwise the test is invalid — you blocked SHH but the barrier didn't change).
FAIL if	SHH-blocked tumor is smaller than or equal to control (the paradox didn't emerge), OR barrier density is unchanged (SHH pathway has no effect on stroma, wiring is disconnected).

Anchor 4: SMAD4-Intact TGF-β Brakes the Tumor

Metrics

ID	Metric	How to measure
4.1	**Tumor cell count**	Viable tumor cells at endpoint
4.2	**Mean tumor proliferation rate**	Average cycle.transition_rate across all tumor cells at endpoint
4.3	**Barrier density**	Mean peri-tumoral ECM density

Compare: **SMAD4-null** (tgfb_brake_sensitivity = 0.05) vs. **SMAD4-intact** (tgfb_brake_sensitivity = 0.8). All other parameters identical.

Pass/Fail

Test	Criterion
P4a	Tumor cell count (SMAD4-intact) < Tumor cell count (SMAD4-null). TGF-β is slowing the SMAD4-intact tumor.
P4b	Mean proliferation rate (SMAD4-intact) < Mean proliferation rate (SMAD4-null). The brake is mechanistically engaging, not just a downstream consequence.
P4c	Barrier density (SMAD4-intact) ≤ Barrier density (SMAD4-null). Slower tumor → less paracrine pressure → equal or thinner stroma. An *increase* in barrier density would be acceptable only if explainable by prolonged TGF-β exposure from the slower-growing tumor, but should raise a flag for investigation.
FAIL if	SMAD4-intact tumor grows as fast as or faster than SMAD4-null. This means TGF-β is not functioning as a brake even when SMAD4 is present — the dual-nature wiring is broken.

Anchor 5: Hypoxia Develops Centrally as Barrier Matures

Metrics

ID	Metric	How to measure
5.1	**O_2 at tumor centroid**	Oxygen substrate concentration at geometric center of tumor
5.2	**O_2 at tumor periphery**	Mean oxygen concentration in cells within 50μm of the tumor-stroma boundary
5.3	**HIF1A activation fraction (core vs. periphery)**	Fraction of HIF1A=ON cells in inner 50% of tumor radius vs. outer 50%

Measure at early, mid, and late time points.

Pass/Fail

Test	Criterion
P5a	O_2 at centroid (late) < O_2 at centroid (early). Central oxygenation declines as barrier matures.
P5b	O_2 at centroid (late) < O_2 at periphery (late). A spatial gradient exists at the mature stage — center is more hypoxic than edge.
P5c	HIF1A activation fraction (core) > HIF1A activation fraction (periphery) at late time point. More cells in the interior have activated HIF1A than at the surface.
FAIL if	O_2 is uniform across the tumor at all time points (no gradient forms — check ECM→O_2 diffusion coupling and domain size), or if HIF1A activation is higher at the periphery than the core (gradient is inverted — check O_2 boundary conditions).

Anchor 6: EMT Is Spatially Restricted to the Periphery

Metrics

ID	Metric	How to measure
6.1	**EMT fraction (core)**	Fraction of tumor cells with ZEB1=ON in the inner 50% of tumor radius
6.2	**EMT fraction (periphery)**	Same, in the outer 50% of tumor radius
6.3	**CDH1 expression gradient**	Mean CDH1 state (ON=1, OFF=0) as a function of distance from centroid

Pass/Fail

Test	Criterion
P6a	EMT fraction (periphery) > EMT fraction (core). More mesenchymal cells at the edge than in the center.
P6b	EMT fraction (core) < 0.5 (majority of core cells remain epithelial). The whole tumor should NOT go mesenchymal.
P6c	CDH1 expression gradient is positive (higher CDH1 / more epithelial toward center, lower / more mesenchymal toward edge). The gradient need not be perfectly smooth, but the trend must be present.
FAIL if	EMT is uniform (threshold too low or TGF-β flooding the entire domain), or EMT fraction is zero everywhere (threshold too high or ZEB1 activation disconnected), or EMT is paradoxically higher in the core than at the periphery (signal gradients are inverted).

Anchor 7: ECM Degradation Improves Drug Delivery But Is Insufficient Alone

Metrics

ID	Metric	How to measure
7.1	**Tumor cell viability**	Fraction of tumor cells alive at endpoint
7.2	**Drug at centroid**	Drug substrate concentration at tumor center after standard diffusion period
7.3	**ECM recovery rate**	Slope of mean ECM density vs. time after degradation event (positive = rebuilding)

Compare three arms: **Control** (no intervention), **Degrade-only** (ECM reduction, no drug), **Degrade+Drug** (ECM reduction + drug from boundary).

Pass/Fail

Test	Criterion
P7a	Viability (Degrade-only) ≥ Viability (Control). ECM removal without drug does not kill tumor — may even allow expansion.
P7b	Drug at centroid (Degrade+Drug) > Drug at centroid (Control with drug). Degradation actually improves drug delivery.
P7c	Viability (Degrade+Drug) < Viability (Control with drug). The combination is better than drug alone against an intact barrier.
P7d	ECM recovery rate (Degrade-only) > 0. After degradation, ECM begins rebuilding. The barrier is not permanently destroyed.
P7e	Viability (Degrade+Drug) > 0. Some tumor cells survive even the combination — complete eradication should NOT occur (if it does, efflux resistance or spatial sanctuary is too weak).
FAIL if	ECM degradation doesn't improve drug penetration (P7b fails — coupling is broken), or barrier doesn't rebuild (P7d fails — CAFs are dead or disconnected), or degrade-only kills the tumor (P7a fails — something is producing cell death without drug).

Anchor 8: HA and Collagen Are Distinct Barrier Components

Metrics

ID	Metric	How to measure
8.1	**Drug penetration half-depth**	Distance from boundary where drug = 50% of boundary concentration

ID	Metric	How to measure
8.2	**Tumor spatial extent**	Maximum radius of tumor cell population
8.3	**Interstitial pressure proxy**	Mean ECM density in the peri-tumoral shell (HA-dominated component if tracked separately)

Compare three arms from mature barrier: **HA-depleted** (HAS2=0), **Collagen-depleted** (COL1A1=0), **Both-depleted**.

Pass/Fail

Test	Criterion
P8a	The *rank order* of drug penetration half-depth differs between HA-depleted and Collagen-depleted arms. They should NOT produce identical improvements — if they do, they're functionally redundant in the model.
P8b	Tumor spatial extent (Collagen-depleted) > Tumor spatial extent (HA-depleted). Collagen provides more structural confinement than HA; removing collagen releases the tumor more. (Direction may be debatable — flag for literature review if it reverses, but some asymmetry must exist.)
P8c	Drug penetration (Both-depleted) > Drug penetration (either single depletion). Removing both components improves access more than either alone. Some additivity or synergy.
P8d	Drug penetration (Both-depleted) > max(Drug penetration HA-depleted, Drug penetration Collagen-depleted). The combined effect exceeds the best single-component depletion — not just equal to it.
FAIL if	HA-depleted and Collagen-depleted produce statistically identical effects across all three metrics (components are not mechanistically distinct — they're just two names for the same ECM field), or if removing both shows no improvement over removing one (no additivity — one component is doing all the work).

Anchor 9: Thicker Barrier Predicts Worse Drug Response

Metrics

ID	Metric	How to measure
9.1	**Pre-treatment barrier density**	Mean peri-tumoral ECM density at the moment drug is introduced
9.2	**Post-treatment tumor viability**	Fraction of tumor cells alive after a fixed drug exposure period
9.3	**Spearman rank correlation**	Correlation between 9.1 and 9.2 across the PSC density sweep (low/medium/high initial seeding)

Run identical tumor genotype + identical drug regimen across ≥3 PSC seeding densities.

Pass/Fail

Test	Criterion
P9a	Viability (high PSC density) > Viability (medium) > Viability (low). Strict monotonic ordering. More stroma → more surviving tumor, same drug, same tumor genetics.
P9b	Pre-treatment barrier density (high PSC) > (medium) > (low). Confirms the PSC density sweep actually produced different barrier thicknesses (otherwise the test is invalid).
P9c	Spearman rank correlation between barrier density and viability is positive. This is equivalent to P9a but stated as a correlation for clarity when running larger sweeps.

Test	Criterion
FAIL if	Viability is independent of PSC density (drug efficacy doesn't depend on barrier — ECM→diffusion coupling is broken), or the relationship is non-monotonic without explanation (middle density worse than high — investigate for interaction effects but treat as likely bug).

Anchor 10: Survivors Cluster in the Spatial Sanctuary

Metrics

ID	Metric	How to measure
10.1	**Survivor spatial concentration**	Mean ECM density at the locations of surviving tumor cells vs. mean ECM density at the locations of dead tumor cells
10.2	**Survivor distance from boundary**	Mean distance of surviving cells from the drug source (domain boundary) vs. mean distance of dead cells
10.3	**Regrowth after drug withdrawal**	Tumor cell count at time T after drug removal vs. tumor cell count at moment of drug removal
10.4	**Barrier re-emergence**	Mean peri-tumoral ECM density at time T after drug removal vs. at moment of drug removal

Pass/Fail

Test	Criterion
P10a	Mean ECM at survivor locations > Mean ECM at dead cell locations. Survivors are behind thicker barrier, not randomly distributed.

Test	Criterion
P10b	Mean distance from boundary of survivors > Mean distance of dead cells. Survivors are deeper in the tumor, farther from drug source.
P10c	P10a and P10b must *both* pass. Spatial sanctuary requires both ECM protection AND distance protection. If only one passes, the sanctuary effect is incomplete.
P10d	Tumor cell count (post-withdrawal T) > Tumor cell count (at drug removal). The sanctuary population regrows.
P10e	ECM density (post-withdrawal T) > ECM density (at drug removal). The barrier rebuilds around the regrowing population.
P10f	The regrowing population re-establishes paracrine signaling: CAF activation fraction post-regrowth > CAF activation fraction at drug removal. The full cycle (tumor → stroma → barrier → sanctuary) reinitializes.
FAIL if	Survivors are randomly located (P10a/b fail — drug penetration is uniform, barrier has no spatial effect), or survivors don't regrow (P10d fails — surviving cells lost proliferative capacity, check genotype preservation), or barrier doesn't rebuild (P10e/f fail — CAF activation loop is broken in the post-treatment state).

Master Test Matrix

Anchor	# Tests	Key comparison	Minimum replicate runs
1	3	Before vs. after (same run)	3 (stochastic seeding)
2	3	Early vs. mid vs. late (same run)	3
3	3	Control vs. SHH-blocked	3 per arm (6 total)

Anchor	# Tests	Key comparison	Minimum replicate runs
4	3	SMAD4-null vs. SMAD4-intact	3 per arm (6 total)
5	3	Early vs. late + core vs. periphery	3
6	3	Core vs. periphery (same run)	3
7	5	Control vs. Degrade vs. Degrade+Drug	3 per arm (9 total)
8	4	HA-depleted vs. Collagen-depleted vs. Both	3 per arm (9 total)
9	3	Low vs. Med vs. High PSC density	3 per condition (9 total)
10	6	Spatial survivor analysis + regrowth	3

Total minimum runs for full anchor validation: ~60 simulation runs.

At estimated wall-time of [TBD] per run on LARCC, this is the HPC budget required before any EA evolution begins. Worth it — every bug caught here saves hundreds of wasted EA generations downstream.

Decision Protocol

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For each anchor:
  1. Run ≥3 replicates per arm (different random seeds)
  2. Compute median metric values across replicates
  3. Check all pass/fail criteria against medians
  4. If ALL tests pass → Anchor PASSED → proceed to next
  5. If ANY test fails:
     a. Check "FAIL if" interpretation → identify suspect subsystem
     b. Fix the suspect wiring / coupling / parameter
     c. Re-run THIS anchor AND all anchors it depends on
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d. Do not proceed to dependent anchors until parent passes

Anchor 1 must pass before any other anchor is attempted.
Anchor 10 is attempted only after Anchors 1–9 all pass.
Anchor 10 passing = model is validated for EA deployment.

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34 total pass/fail tests across 10 anchors. Every one is directional: A > B, A < B, or correlation is positive. No magic numbers. If the biology is qualitatively right, these all pass. If any fails, the failure interpretation tells you where to look.